

MODULE 5

UNIT 5.1: FEEDBACK AMPLIFIERS



Outline



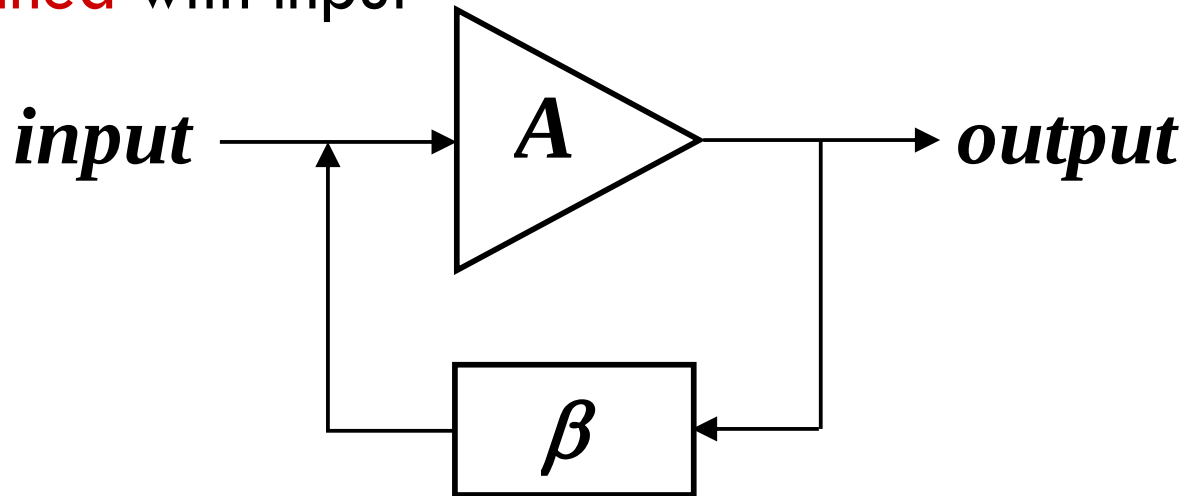
1. Introduction to Feedback
2. Feedback Amplifier – Positive & Negative
3. Advantages/Disadvantages of Negative Feedback
4. Basic Feedback Concept
5. Classification of Amplifiers

Introduction to Feedback

- Feedback is used in virtually all amplifier system.
- Invented in 1928 by Harold Black – engineer in Western Electric Company
 - ▣ methods to stabilize the gain of amplifier for use in telephone repeaters.
- In feedback system, a signal that is proportional to the output is fed back to the input and combined with the input signal to produce a desired system response.
- However, unintentional and undesired system response may be produced.

Feedback Amplifier

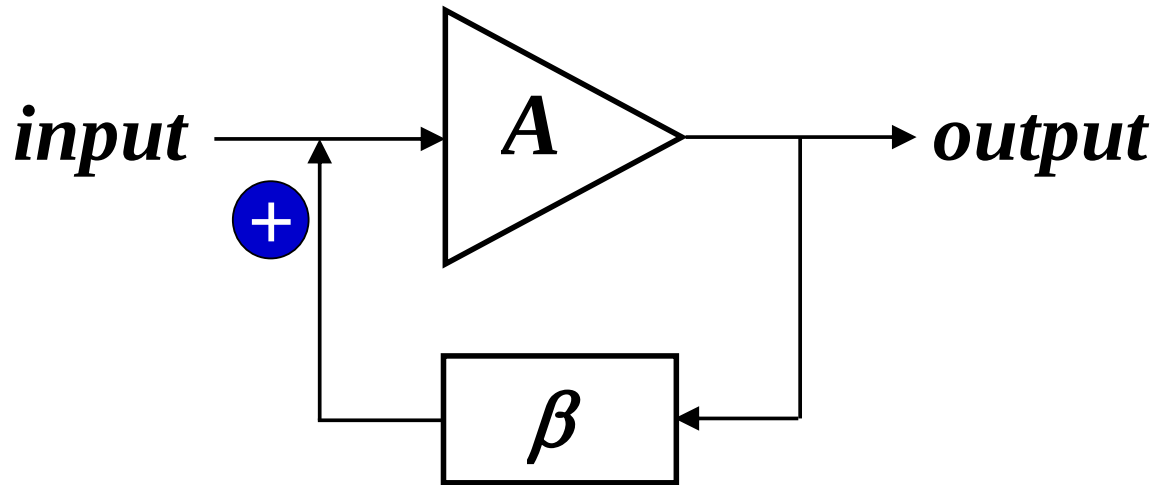
- *Feedback* is a technique where a proportion of the output of a system (amplifier) is **fed back** and **recombined** with input



- There are 2 types of feedback amplifier:
 - ▣ Positive feedback
 - ▣ Negative feedback

Positive Feedback

- Positive feedback is the process when the output is *added* to the input, amplified again, and this process continues.



- Positive feedback is used in the design of oscillator and other application.

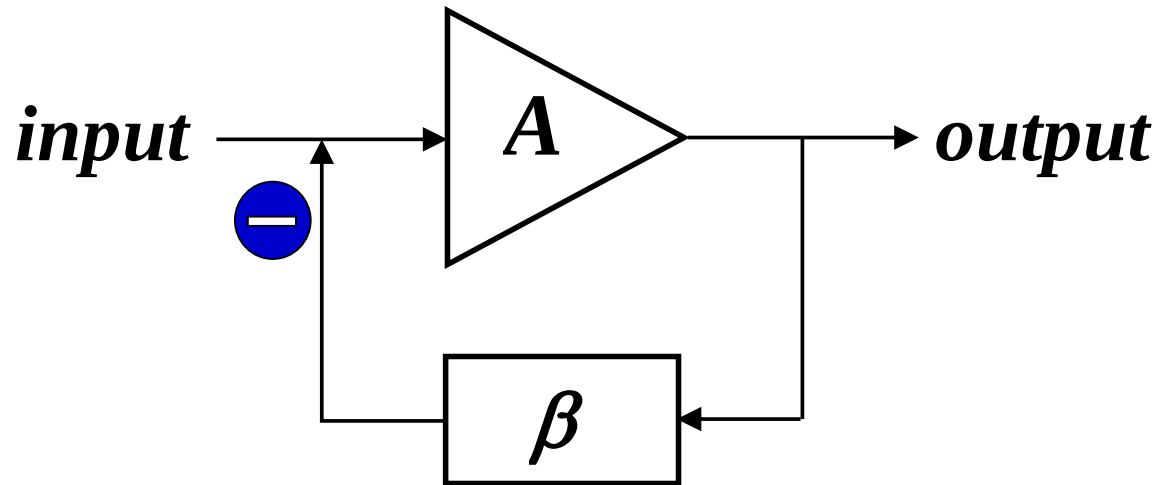
Positive Feedback - Example

- In a PA system

get feedback when you put the microphone in front of a speaker and the sound gets uncontrollably loud (you have probably heard this unpleasant effect).

Negative Feedback

- Negative feedback is when the output is **subtracted** from the input.



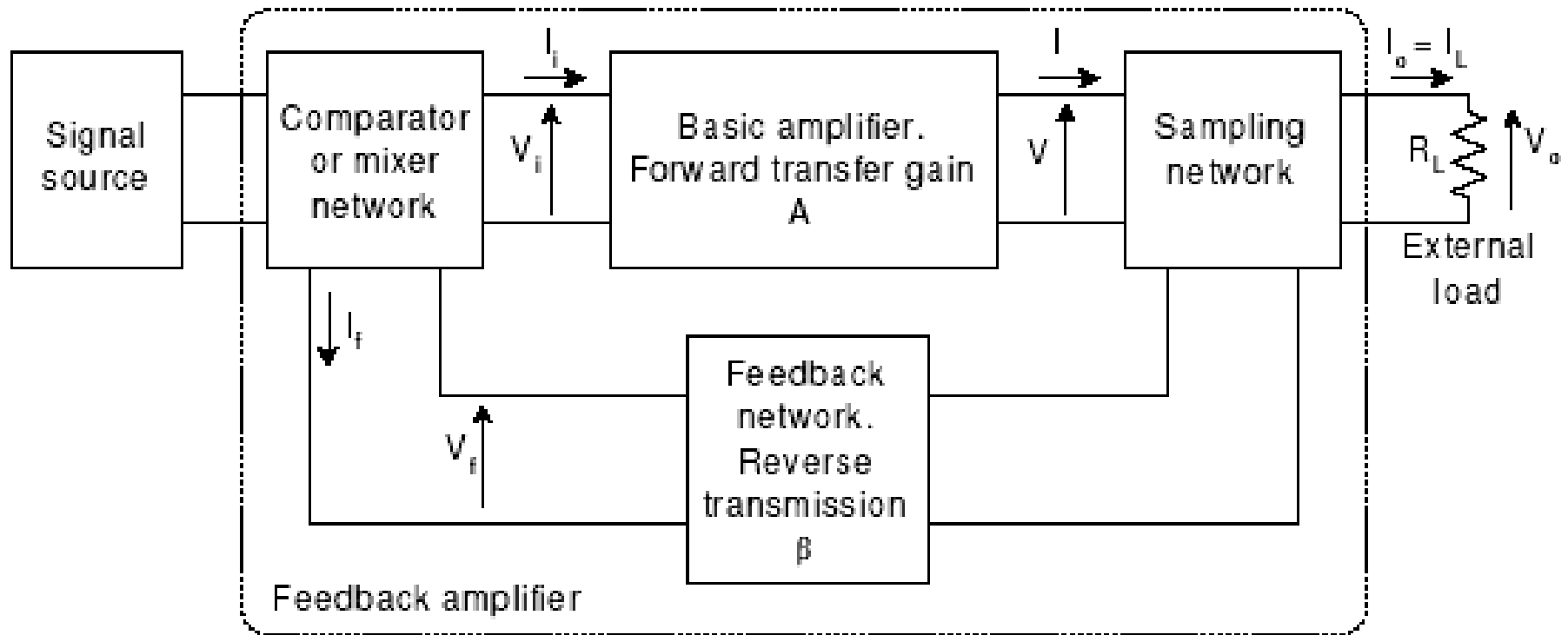
- The use of negative feedback reduces the gain. Part of the output signal is taken back to the input with a negative sign.

Negative Feedback - Example

□ Speed control

If the car starts to speed up above the desired set-point speed, negative feedback causes the throttle to close, thereby reducing speed; similarly, if the car slows, negative feedback acts to open the throttle

Feedback Amplifier - Concept



Basic structure of a single - loop feedback amplifier

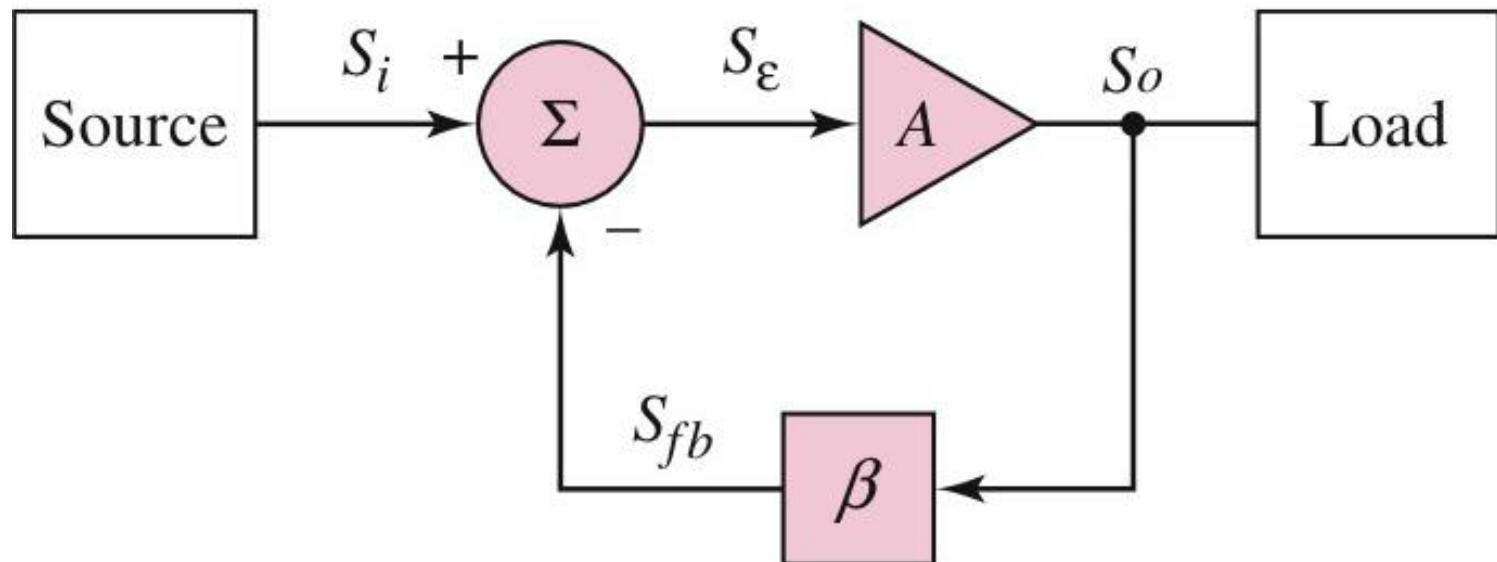
Advantages of Negative Feedback

1. Gain Sensitivity – variations in gain is reduced.
2. Bandwidth Extension – larger than that of basic amplified.
3. Noise Sensitivity – may increase S-N ratio.
4. Reduction of Nonlinear Distortion
5. Control of Impedance Levels – input and output impedances can be increased or decreased.

Disadvantages of Negative Feedback

1. Circuit Gain – overall amplifier gain is reduced compared to that of basic amplifier.
2. Stability – possibility that feedback circuit will become unstable and oscillate at high frequencies.

Basic Feedback Concept



Basic configuration of a feedback amplifier

Basic Feedback Concept

- The output signal is: $S_o = AS_\varepsilon$
where A is the amplification factor
- Feedback signal is $S_{fb} = \beta S_o$
where β is the feedback transfer function
- At summing node: $S_\varepsilon = S_i - S_{fb}$
- Closed-loop transfer function or gain is

$$A_f = \frac{S_o}{S_i} = \frac{A}{1 + \beta A}$$

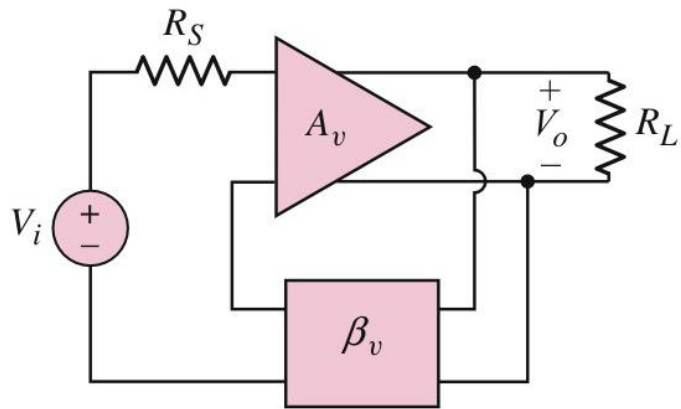
if $\beta A \gg 1$ then $A_f \cong \frac{A}{\beta A} = \frac{1}{\beta}$

Classification of Amplifiers

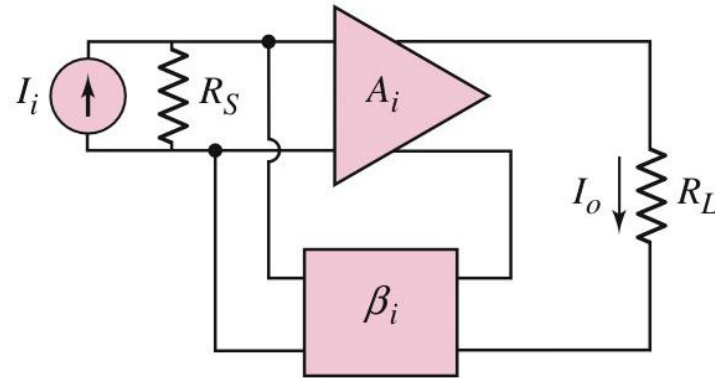
Classify amplifiers based on their input (parameter to be amplified; voltage or current) & output signal relationships:

- Voltage amplifier---series-shunt feedback
voltage mixing and voltage sampling
- Current amplifier---shunt-series feedback
Current mixing and current sampling
- Transconductance amplifier---series-series feedback
Voltage mixing and current sampling
- Transresistance amplifier---shunt-shunt feedback
Current mixing and voltage sampling

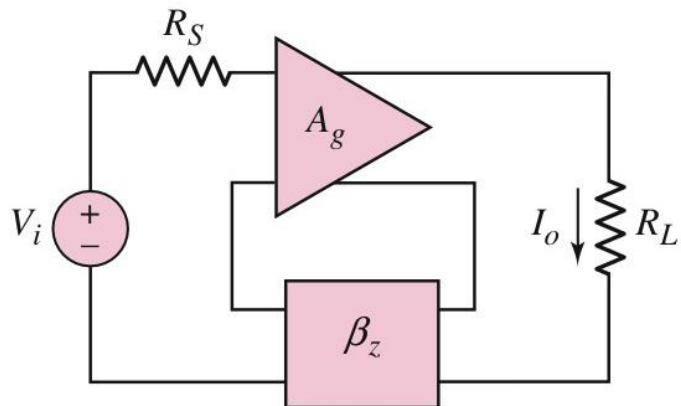
Feedback Configuration



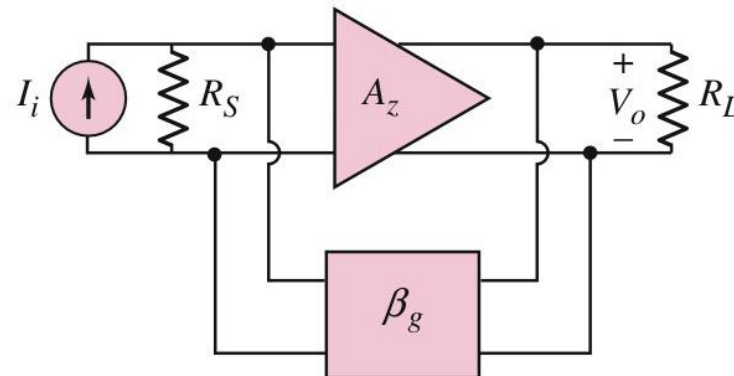
(a) Series-shunt



(b) Shunt-series



(c) Series-series



(d) Shunt-shunt

Series:

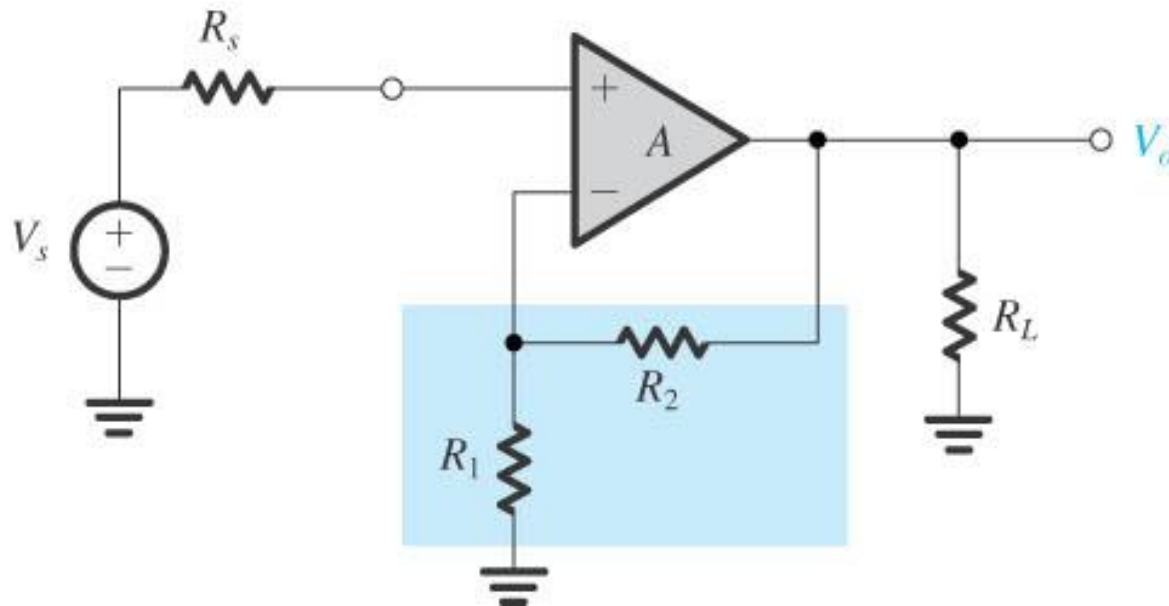
connecting the feedback signal in series with the input signal voltage.

Shunt:

connecting the feedback signal in shunt (parallel) with an input current source

The Amplifier with Series-Shunt Feedback

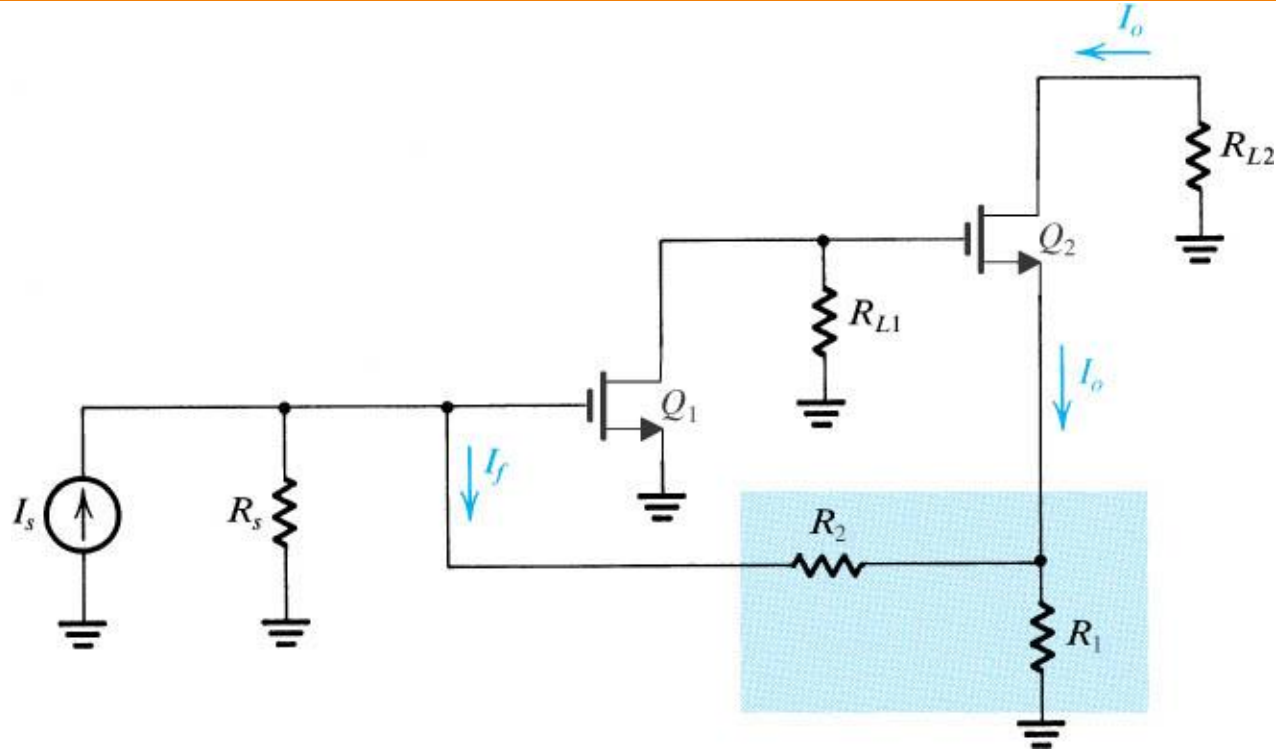
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voltage-mixing voltage-sampling (series–shunt) topology

The Amplifier with Shunt-Series Feedback

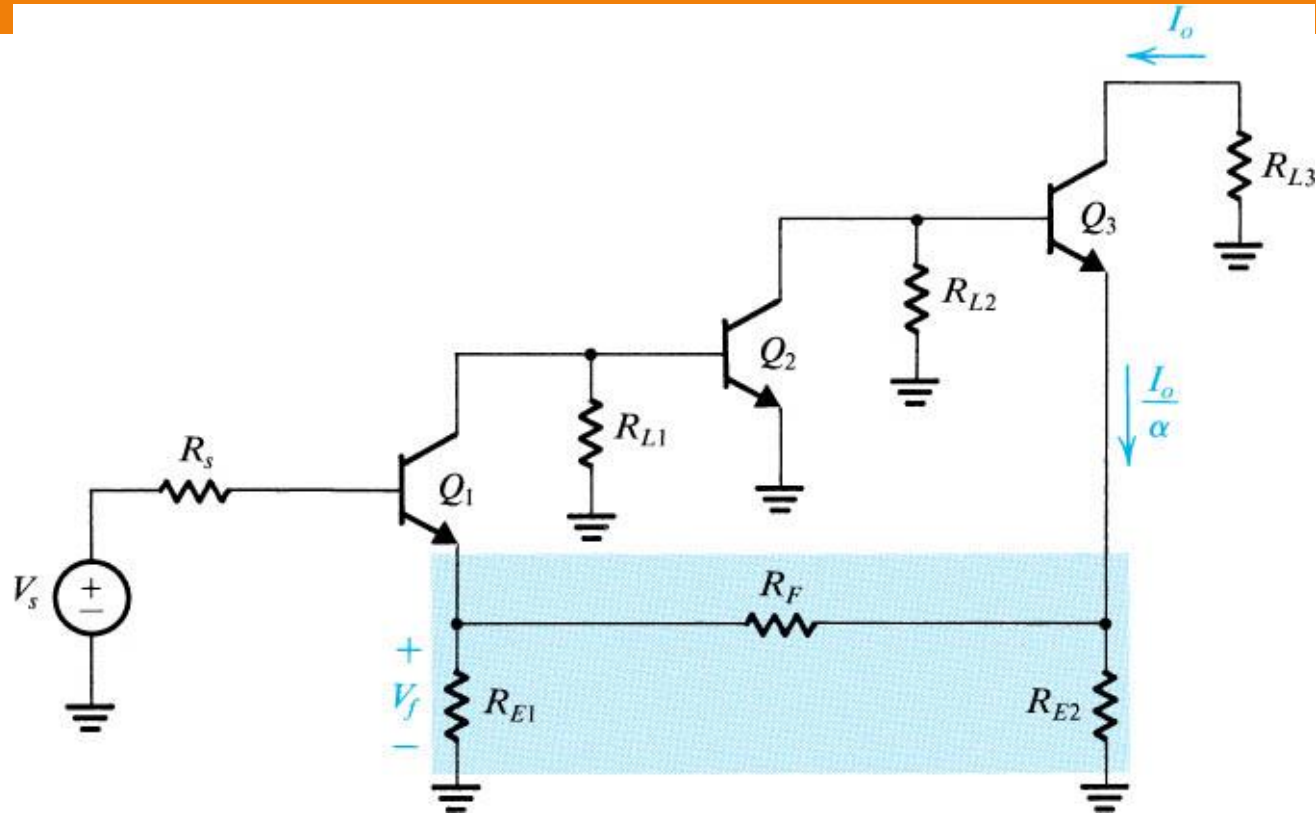
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current-mixing current-sampling (shunt-series) topology

The Amplifier with Series-Series Feedback

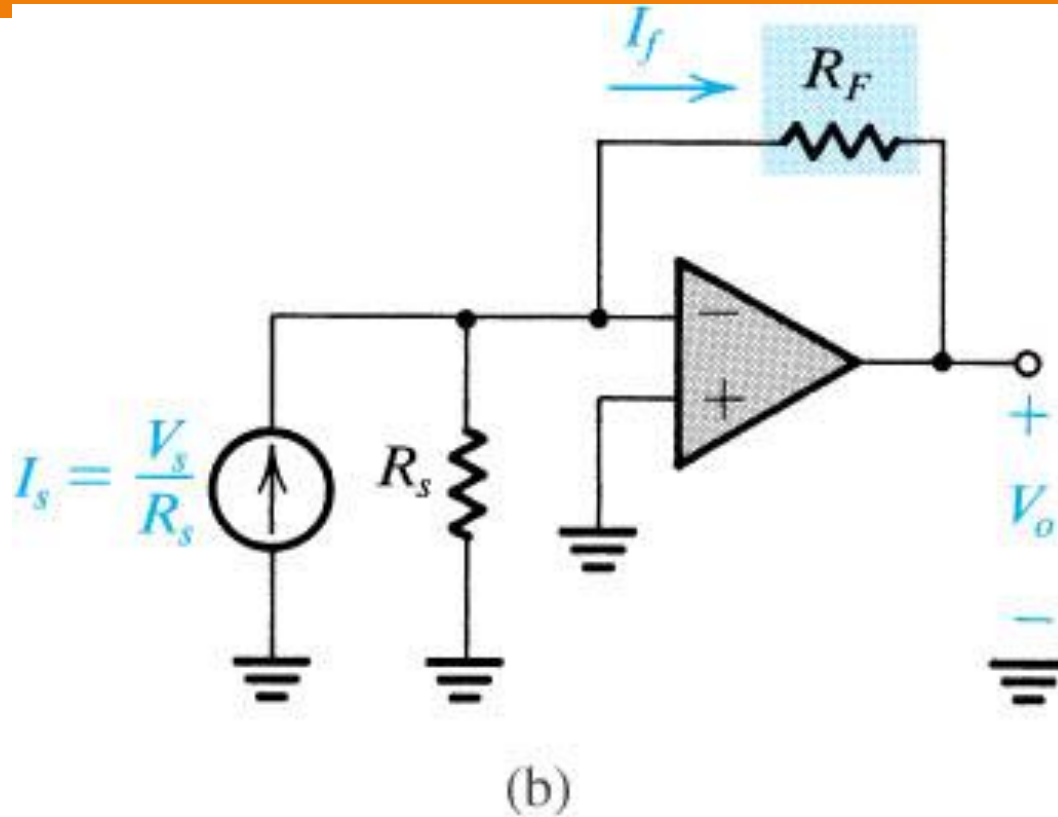
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voltage-mixing current-sampling (series-series) topology

The OP Amplifier with Shunt-Shunt Feedback

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current-mixing voltage-sampling (shunt–shunt) topology

Feedback Amplifier

Input and output Impedances

□ Summary

1. For a series connection at input or output, the resistance is increased by $(1 + \beta A)$.
2. For a shunt connection at input or output, the resistance is lowered by $(1 + \beta A)$.

Feedback Amplifier

Feedback amplifier	Source signal	Output signal	Transfer function	Input Resistance	Output Resistance
Series-Shunt (voltage amplifier)	Voltage	Voltage	$A_{vf} = \frac{A_v}{1 + \beta_v A_v}$	$(1 + \beta_v A_v) R_i$	$\frac{R_o}{(1 + \beta_v A_v)}$
Shunt-Series (current amplifier)	Current	Current	$A_{if} = \frac{A_i}{1 + \beta_i A_i}$	$\frac{R_i}{(1 + \beta_i A_i)}$	$(1 + \beta_i A_i) R_o$
Series-Series (transconductance amplifier)	Voltage	Current	$A_{gf} = \frac{A_g}{1 + \beta_g A_g}$	$(1 + \beta_g A_g) R_i$	$(1 + \beta_g A_g) R_o$
Shunt-Shunt (transresistance amplifier)	Current	Voltage	$A_{zf} = \frac{A_z}{1 + \beta_z A_z}$	$\frac{R_i}{(1 + \beta_z A_z)}$	$\frac{R_o}{(1 + \beta_z A_z)}$